# PHYSICS
# CLASS 10

## 132 Summative Theory SLOs with Exam-Based Answers

FBISE SSC-II | Sendup Examination Preparation

## Made by Fakhar STEM Sphere

Definitions • Reasoning • Comparison tables
Formulas • Worked examples • Exam diagrams

### Scope

All 132 theory-assessed SLOs in the supplied 15-page assessment framework: 28 Heat and Thermodynamics, 2 Weather, 52 Waves/Sound/Optics, and 50 Electricity SLOs.

Includes the three SLOs labelled Theory and PBA. Formative and PBA-only rows are excluded.

Source coverage ends at P-10-E-59. This is a study resource based on the supplied framework, not an official FBISE answer key.

# Contents and study guidance

1. Heat and thermodynamics 3

2. Weather and climate 12

3. Waves 13

4. Sound 17

5. Geometrical optics 22

6. Electrostatics 28

7. Current electricity and circuits 33

8. Electrical appliances and mains 41

Appendix A. Lens and eye ray diagrams 44

Appendix B. Current-voltage graphs 46

Appendix C. Circuit symbols 47

## How to write an exam answer

For a definition: state the essential scientific meaning, then add the required formula and SI unit. For an explanation: give the physical cause, the process and the result. For a numerical: write Given → Formula → Substitution → Answer with unit.

For comparisons, use parallel points in a table. For diagrams, draw neat rays or wires, add directions and label the important parts. Match answer length to the marks and command words in the question.

## Scientific notes retained from the answers

Tsunami frequency remains approximately constant as the wave enters shallow water; speed and wavelength decrease while height generally rises. This corrects the wording in SLO P-10-D-10.

The source prints the insulator-discharge SLO as P-09-E-05; that identifier is retained here.

Navigation: the PDF bookmarks list each section and all 132 SLOs. Diagram appendices are referenced from relevant answers.

# 1. Heat and thermodynamics

## 1. SLO P-10-C-01 - Define and calculate specific heat.

Answer

Specific heat capacity is the thermal energy required to raise the temperature of 1 kg of a substance by 1 K, without changing its state.

$$c = Q / (m\Delta T) \text{ or } Q = mc\Delta T$$

Its SI unit is J $kg^{-1}$ $K^{-1}$.

**Worked example:** If 2 kg of a substance absorbs 8,000 J and its temperature rises by 10 K:

$$c = 8000 / (2 \times 10) = 400 \text{ J } kg^{-1} K^{-1}$$

## 2. SLO P-10-C-03 - Analyze everyday effects due to the large specific heat of water.

Answer

Water has a high specific heat capacity, approximately 4200 J $kg^{-1}$ $K^{-1}$. Therefore, it absorbs or releases considerable thermal energy with a relatively small temperature change.

- **Moderate coastal climates:** The sea heats and cools slowly, reducing temperature extremes near coasts.
- **Engine cooling:** Water absorbs considerable heat from an engine.
- **Hot-water bottles:** Water releases a large amount of energy as it cools.
- **Stable aquatic temperatures:** Slow temperature changes help aquatic organisms survive.

## 3. SLO P-10-C-04 - Use the terms for changes of state between solids, liquids and gases, including deposition and sublimation.

Answer

| Change of state | Term | Example |
|---|---|---|
| Solid → liquid | Melting | Ice changing into water |
| Liquid → solid | Freezing or solidification | Water changing into ice |
| Liquid → gas | Vaporization | Water changing into steam |
| Gas → liquid | Condensation | Droplets on a cold glass |
| Solid → gas directly | Sublimation | Dry ice changing into gas |
| Gas → solid directly | Deposition | Water vapour forming frost |

Melting, vaporization and sublimation absorb energy. Freezing, condensation and deposition release energy.

## 4. SLO P-10-C-05 - Explain thermal expansion using kinetic theory and state the relative expansion of solids, liquids and gases.

Answer

Thermal expansion is the increase in a substance's dimensions when its temperature rises.

- **Solids:** Heating increases particle vibrations. Their average separation usually increases slightly, causing expansion.
- **Liquids:** Particles gain energy and move more vigorously. Their average separation increases, producing greater expansion than in most solids.
- **Gases:** At constant pressure, faster-moving particles occupy a larger volume.

For comparable temperature rises, the usual order is:

> Expansion of gases > liquids > solids

**Important:** The particles themselves do not become larger; their average separation increases.

## 5. SLO P-10-C-06 - Analyze applications and consequences of thermal expansion in real life.

Answer

| Application or consequence | Explanation |
|---|---|
| Liquid-in-glass thermometer | The liquid expands when heated and rises in the narrow tube. |
| Bimetallic thermostat | Joined metals expand differently, bending the strip and operating a switch. |
| Railway tracks and bridges | Expansion gaps or joints allow expansion and prevent buckling. |
| Overhead cables | Suitable sag accommodates temperature changes. |
| Tight metal jar lid | Warming expands the lid, making it easier to remove. |
| Hot riveting | A heated rivet contracts on cooling and holds plates tightly. |

## 6. SLO P-10-C-07 - Analyze melting, solidification, boiling and condensation in terms of energy transfer without temperature change.

Answer

During a change of state of a pure substance at constant pressure, energy is absorbed or released while its temperature remains constant.

| Process | Energy transfer | Particle explanation |
|---|---|---|
| Melting | Energy absorbed | The ordered solid structure breaks down as particles gain freedom of movement. |
| Solidification | Energy released | Particles become arranged in a more ordered solid structure. |

| Process | Energy transfer | Particle explanation |
|---|---|---|
| Boiling | Energy absorbed | Particles separate sufficiently to form a gas. |
| Condensation | Energy released | Gas particles come closer and form a liquid. |

The energy changes particle potential energy and arrangement rather than increasing average kinetic energy. Therefore, the temperature remains constant during the transition.

## 7. SLO P-10-C-08 - State the melting and boiling temperatures of water at standard atmospheric pressure.

Answer

At standard atmospheric pressure, approximately $1.013 \times 10^5$ Pa:

Melting point of ice = 0°C = 273.15 K

Boiling point of water = 100°C = 373.15 K

The boiling point changes when atmospheric pressure changes.

## 8. SLO P-10-C-09 - Describe thermal expansion of solids: linear and volumetric expansion.

Answer

**Linear expansion** is an increase in the length of a solid on heating.

$$\Delta L = \alpha L_o \Delta T \text{ and } L = L_o(1 + \alpha \Delta T)$$

**Volumetric expansion** is an increase in the volume of a solid on heating.

$$\Delta V = \beta V_o \Delta T \text{ and } V = V_o(1 + \beta \Delta T)$$

Here α and β are the coefficients of linear and volume expansion. Both have SI unit $K^{-1}$. For an isotropic solid and small expansion:

$$\beta \approx 3\alpha$$

## 9. SLO P-10-C-10 - Explain real and apparent expansion of liquids.

Answer

When a liquid is heated in a vessel, both the liquid and the vessel expand.

- **Real expansion:** The actual increase in the volume of the liquid.
- **Apparent expansion:** The increase observed relative to the expanding vessel.

Because the vessel also becomes larger:

$$\Delta V(\text{real}) = \Delta V(\text{apparent}) + \Delta V(\text{vessel})$$

For small expansion:

$$\beta(\text{real}) = \beta(\text{apparent}) + \beta(\text{vessel})$$

Thus apparent expansion is normally smaller than real expansion. When a flask is first heated, the liquid level may initially fall because the flask expands before the liquid becomes equally warm.

## 10. SLO P-10-C-11 - Explain gas pressure and changes in pressure in terms of particles.

Answer

Gas particles move continuously and randomly. Collisions with the container walls exert forces on them. The force per unit area is pressure:

$$p = F / A$$

- **Heating at constant volume:** Faster particles collide more frequently and forcefully with the walls, increasing pressure.
- **Decreasing volume at constant temperature:** Collisions with walls become more frequent, increasing pressure.
- **Adding more gas at constant volume and temperature:** More particles produce more collisions, increasing pressure.

## 11. SLO P-10-C-12 - Differentiate between boiling and evaporation.

Answer

| Evaporation | Boiling |
|---|---|
| Occurs at the liquid's surface | Occurs throughout the liquid |
| Can occur below the boiling point | Occurs at the boiling temperature for the given pressure |
| Usually slow | Usually rapid |
| No vapour bubbles form throughout the liquid | Vapour bubbles form throughout the liquid |
| Produces cooling | During steady boiling, supplied energy changes state without raising temperature |

## 12. SLO P-10-C-22 - Describe evaporation in terms of the escape of more energetic particles from a liquid's surface.

Answer

Liquid particles have a range of kinetic energies. Some near the surface have enough kinetic energy to overcome the attractive forces holding them in the liquid. They escape into the surrounding space as vapour.

This process is evaporation. It occurs at the surface and can take place below the boiling point.

## 13. SLO P-10-C-23 - Explain the effects of temperature, humidity, surface area and air movement on evaporation.

Answer

| Factor | Effect | Reason |
|---|---|---|
| Higher temperature | Increases evaporation | More particles have enough energy to escape. |
| Higher humidity | Decreases net evaporation | The surrounding air already contains more water vapour. |
| Larger surface area | Increases evaporation | More particles are exposed at the surface. |
| Greater air movement | Increases evaporation | Moving air removes vapour above the surface. |

Therefore, clothes dry faster on a warm, dry and windy day when spread out.

## 14. SLO P-10-C-24 - Explain how evaporation causes cooling.

Answer

During evaporation, the more energetic particles escape from a liquid. The average kinetic energy of the remaining particles decreases. Since temperature depends on average kinetic energy, the liquid's temperature falls.

The liquid may then absorb thermal energy from its surroundings, cooling them.

**Example:** Sweat evaporates from the skin by absorbing energy from the body, producing cooling.

## 15. SLO P-10-C-25 - Describe cooling by evaporation in refrigeration without harmful CFCs.

Answer

1. A liquid refrigerant enters the evaporator at low pressure.

2. It evaporates and absorbs latent heat from the refrigerator's interior.

3. A compressor compresses the vapour, raising its pressure and temperature.

4. In the condenser outside the cabinet, the refrigerant releases energy and condenses.

5. An expansion device lowers its pressure, and the cycle repeats.

The interior is cooled because refrigerant evaporation absorbs energy. CFC-free refrigerants, such as carbon dioxide or suitable hydrocarbons, can replace ozone-depleting CFCs.

## 16. SLO P-10-C-26 - Explain latent heat in terms of energy, particle behaviour and forces between particles.

Answer

Latent heat is the energy absorbed or released when a substance changes state without a temperature change.

Specific latent heat is the energy required to change the state of 1 kg of a substance:

$$L = Q / m \text{ or } Q = mL$$

Its SI unit is J $kg^{-1}$.

- **Specific latent heat of fusion:** Energy per kilogram needed to change solid into liquid at its melting point.
- **Specific latent heat of vaporization:** Energy per kilogram needed to change liquid into gas at its boiling point.

The energy changes particle separation and arrangement against attractive forces rather than raising temperature.

## 17. SLO P-10-C-30 - Justify experiments to distinguish good and bad thermal conductors.

Answer

Take rods of different materials but equal length and cross-sectional area. Attach small drawing pins with equal amounts of wax at equal distances from the heated ends.

1. Heat one end of each rod equally, for example in the same hot-water bath.

2. Observe the time taken for the wax to melt and the pins to fall.

3. Repeat with the same initial temperature and heating conditions.

**Observation:** Pins fall sooner from good conductors because thermal energy reaches the wax faster.

**Conclusion:** A material transferring thermal energy more rapidly is a better thermal conductor.

## 18. SLO P-10-C-31 - Explain conduction through lattice vibrations and free electrons in metals.

Answer

Conduction is thermal energy transfer through a substance without bulk movement of that substance.

In solids, particles at the hot end vibrate more vigorously. Through interactions with neighbours, energy passes through the lattice towards colder regions.

Metals also contain free, delocalized electrons. These transfer energy rapidly by moving through the metal and interacting with the lattice and other electrons.

Therefore, metals are generally better thermal conductors than non-metallic solids.

## 19. SLO P-10-C-32 - Explain convection using density changes and describe experiments illustrating it.

Answer

Convection is thermal energy transfer by the bulk movement of a liquid or gas.

When a fluid is heated, it usually expands, becomes less dense and rises. Cooler, denser fluid moves down to replace it. This circulation is a convection current.

**Water experiment:** Place a small potassium permanganate crystal near one side of the bottom of a water-filled beaker. Heat gently beneath it. Coloured water rises above the heated region,

crosses the top and sinks on the cooler side.

**Air example:** Air above a heater rises while cooler room air moves towards it.

## 20. SLO P-10-C-33 - Explain convection in seawater and its support of marine life.

Answer

Seawater density changes with temperature and salinity. Cold, sufficiently dense seawater sinks, while warmer, less dense water rises. This contributes to ocean circulation and mixing.

It supports marine life by carrying oxygen-rich surface water towards deeper regions, redistributing nutrients, and transferring thermal energy to moderate temperatures.

Therefore, circulation helps maintain suitable conditions for marine ecosystems.

## 21. SLO P-10-C-34 - Explain how land and sea breezes moderate coastal climates.

Answer

**Daytime - sea breeze:** Land heats faster than seawater. Air above land warms and rises. Cooler air moves from sea towards land, cooling the coast.

**Night-time - land breeze:** Land cools faster than seawater. Air above the relatively warmer sea rises, and cooler air moves from land towards sea.

Water's high specific heat capacity makes it heat and cool slowly. Together with coastal air circulation, this reduces temperature extremes near the sea.

## 22. SLO P-10-C-35 - Explain how birds and gliders use thermal currents.

Answer

The Sun heats the ground unevenly. Air above warmer ground heats, expands and becomes less dense than its surroundings. It rises in columns called thermals.

Birds and gliders circle within these rising currents. Upward-moving air provides lift and can increase their height without continuous wing flapping or engine power.

The energy originally comes from solar heating of the ground.

## 23. SLO P-10-C-36 - Describe thermal radiation and state that it needs no medium.

Answer

Thermal radiation is energy transfer by electromagnetic waves, mainly infrared radiation at ordinary temperatures.

It does not require particles or a material medium and can travel through a vacuum.

**Example:** Energy from the Sun reaches Earth through space by radiation. All objects above absolute zero emit thermal radiation.

## 24. SLO P-10-C-37 - Explain the effect of surface colour and texture on infrared emission, absorption and reflection.

Answer

| Surface | Absorption | Emission | Reflection |
|---|---|---|---|
| Dull, matt black | Good absorber | Good emitter | Poor reflector |
| Shiny, polished metal | Poor absorber | Poor emitter | Good reflector |

A good absorber is also a good emitter at the same wavelengths.

**Applications:** Matt black solar collectors absorb radiation effectively. Shiny surfaces in vacuum flasks reduce radiation transfer.

## 25. SLO P-10-C-38 - Explain how radiation emission depends on temperature and surface area.

Answer

A higher surface temperature increases the rate of thermal radiation emission. A larger surface area also increases energy emitted per second when temperature and surface properties remain the same.

For fixed emissivity:

$$P(\text{emitted}) \propto AT^4$$

Here T is absolute temperature in kelvin. Therefore, a larger or hotter radiator can emit energy more rapidly. Net energy loss also depends on the surroundings' temperature.

## 26. SLO P-10-C-39 - Describe experiments comparing good and bad emitters and absorbers of infrared radiation.

Answer

**Comparing emitters:** Fill a Leslie cube, whose faces have different finishes, with hot water. Place an infrared detector at the same distance from each face.

**Observation and conclusion:** The matt black face gives a higher reading than polished metal, showing that matt black is a better emitter.

**Comparing absorbers:** Place identical plates, one matt black and one shiny, at equal distances from an infrared source. Use temperature sensors to compare their temperature rises over the same time.

The matt black plate warms faster, showing that it is a better absorber.

## 27. SLO P-10-C-40 - Explain the greenhouse effect and its contribution to global warming.

Answer

1. Much incoming short-wavelength solar radiation passes through the atmosphere and warms Earth's surface.

2. The surface emits longer-wavelength infrared radiation.

3. Greenhouse gases, including water vapour, carbon dioxide and methane, absorb some outgoing infrared radiation.

4. They emit infrared radiation in all directions, including towards the surface.

The natural greenhouse effect keeps Earth warmer than it would otherwise be.

Increasing greenhouse-gas concentrations reduce energy escape to space at a given surface temperature. Earth warms until energy balance is restored, producing an enhanced greenhouse effect and global warming.

## 28. SLO P-10-C-41 - Explain everyday applications of conduction, convection and radiation.

Answer

| Application | Explanation |
|---|---|
| Kitchen pans | Metal bases conduct energy to food; insulating handles reduce transfer to the hand. |
| Room heating | Air near a heater warms and rises; cooler air moves in, producing convection. |
| Infrared thermometer | Detects infrared radiation and estimates surface temperature. |
| Vacuum flask | A vacuum reduces conduction and convection; reflective surfaces reduce radiation; an insulating stopper reduces transfer through the neck. |
| Building insulation | Loft and cavity-wall materials trap air, reducing conduction and convection. |
| Hot-water system | Water heated near the bottom becomes less dense and rises; cooler water sinks to replace it. |

# 2. Weather and climate

## 29. SLO P-10-F-23 - Use convection to explain cyclone formation.

Answer

1. Warm ocean water heats moist air above it.
2. Warm, moist air rises, creating low pressure near the surface.
3. Surrounding air moves towards this low-pressure region.
4. Rising air cools; water vapour condenses and releases latent heat.
5. This energy supports further rising air and strengthens circulation.
6. Earth's rotation causes moving air to turn, producing rotating winds.

Under suitable conditions, the system develops into a tropical cyclone.

## 30. SLO P-10-F-24 - Explain how global warming contributes to extreme weather events.

Answer

| Event | Contribution of global warming |
|---|---|
| Hurricanes or tropical cyclones | Warmer seas can supply more energy and moisture, increasing intensity and rainfall of strong storms. |
| Heat waves | Higher average temperatures make extreme heat more likely and severe. |
| Heavy rainfall and floods | Warmer air can hold more water vapour, allowing heavier rain when conditions favour rainfall. |
| Droughts | Greater evaporation can dry soil and worsen drought in affected regions. |
| Wildfires | Hotter, drier conditions dry vegetation and increase fire risk. |
| Winter storms | Additional atmospheric moisture can increase heavy snowfall where temperatures remain low enough. |

Climate change affects regions differently; it does not make every storm type more frequent everywhere.

# 3. Waves

## 31. SLO P-10-D-01 - Show that waves transfer energy without transferring matter.

Answer

Place a small floating cork on water and generate waves nearby.

**Observation:** Waves travel across the water, while the cork mainly oscillates about its original position rather than travelling with them.

The waves make the cork move, showing energy transfer. However, water particles mainly oscillate locally.

Therefore, waves transfer energy without net transport of the medium in the ideal wave model.

## 32. SLO P-10-D-02 - Describe wave motion using ropes, springs and water waves.

Answer

Wave motion is the propagation of a disturbance that transfers energy from one place to another.

- **Rope:** Moving one end up and down produces a travelling disturbance.
- **Spring:** Pushing and pulling its end creates moving compressions and rarefactions.
- **Water:** A vibrating source produces travelling surface ripples.

The disturbance moves while individual particles oscillate about equilibrium positions.

## 33. SLO P-10-D-03 - Describe the main features of a wave.

Answer

| Term | Definition |
|---|---|
| Wavefront | A line or surface joining points at the same phase of vibration |
| Wavelength, λ | Distance between successive points in the same phase, such as adjacent crests |
| Frequency, f | Number of complete oscillations per second |
| Time period, T | Time for one complete oscillation |
| Crest | Maximum positive displacement |
| Trough | Maximum negative displacement |
| Compression | Region of increased pressure and particle density |
| Rarefaction | Region of decreased pressure and particle density |
| Amplitude, A | Maximum displacement from equilibrium |
| Wave speed, v | Distance travelled by the disturbance per unit time |

$$f = 1 / T \text{ and } v = f\lambda$$

## 34. SLO P-10-D-04 - Define frequency, wavelength and amplitude.

Answer

- **Frequency:** Number of complete oscillations per second. Unit: hertz (Hz).
- **Wavelength:** Shortest distance between successive points in the same phase. Unit: metre (m).
- **Amplitude:** Maximum displacement from equilibrium. Unit: metre (m).

Amplitude is measured from equilibrium to a crest or trough, not from crest to trough.

## 35. SLO P-10-D-05 - Apply the equation v = fλ.

Answer

A wave travels one wavelength during one time period:

$$v = \lambda / T$$

Since f = 1/T:

$$v = f\lambda$$

**Example:** f = 50 Hz and λ = 0.40 m.

$$v = 50 \times 0.40 = 20 \text{ m s}^{-1}$$

Useful rearrangements: f = v/λ and λ = v/f.

## 36. SLO P-10-D-06 - Explain transverse waves and give examples.

Answer

In a transverse wave, vibration is perpendicular to the direction of energy transfer. For a rope wave travelling horizontally, particles may vibrate vertically.

Examples: waves on a stretched string, electromagnetic waves, seismic S-waves, and surface-water waves treated as transverse in the elementary model.

Transverse displacement waves have crests and troughs.

## 37. SLO P-10-D-07 - Explain longitudinal waves and give examples.

Answer

In a longitudinal wave, particles vibrate parallel to the direction of energy transfer.

- **Compressions:** Particles closer together; higher pressure and density.
- **Rarefactions:** Particles farther apart; lower pressure and density.

Examples include sound in air, compression waves in a spring, and seismic P-waves.

## 38. SLO P-10-D-08 - Describe reflection, refraction and diffraction of waves.

Answer

| Phenomenon | Description | Example |
|---|---|---|
| Reflection | A wave returns into its original medium after meeting a boundary. | An echo |
| Refraction | A wave changes speed on entering another medium; direction usually changes at oblique incidence. | Water waves entering shallow water |
| Diffraction | A wave spreads around an edge or after passing through a gap. | Water waves spreading after a narrow opening |

## 39. SLO P-10-D-09 - Describe how wavelength affects diffraction at an edge.

Answer

Diffraction is spreading into the region behind an obstacle's edge. For the same obstacle, longer wavelengths generally show greater diffraction, while shorter wavelengths show less.

It is especially noticeable when wavelength is comparable to obstacle size.

This helps explain why sound travels around everyday obstacles while visible light produces much sharper shadows.

## 40. SLO P-10-D-10 - Explain tsunami generation and changes as a tsunami approaches shallow water.

Answer

A tsunami can be generated when an underwater earthquake, volcanic eruption or landslide suddenly displaces a large volume of water.

In deep water it has a very long wavelength, high speed and often a small wave height.

As it approaches shallow water, its speed decreases, its wavelength decreases, and its wave height generally increases through shoaling.

For shallow-water waves:

$$v \approx \sqrt{gh}$$

**Correction to the source wording:** Frequency remains approximately constant during shoaling. It does not normally increase simply because the water becomes shallower.

## 41. SLO P-10-D-11 - Explain how wavelength and gap size affect diffraction through a gap.

Answer

Diffraction is greatest when gap width is comparable to wavelength.

- A gap much wider than the wavelength produces limited spreading.
- A gap comparable to the wavelength produces strong spreading.
- Increasing wavelength for a fixed gap increases diffraction.
- Decreasing gap width towards the wavelength also increases diffraction.

The spreading depends on the ratio of wavelength to gap width.

# 4. Sound

## 42. SLO P-10-D-12 - Describe the production of sound.

Answer

Sound is produced by vibrating objects. A source pushes and pulls nearby medium particles, creating alternating compressions and rarefactions that travel as sound waves.

Examples include vibrating vocal cords, a tuning fork, a loudspeaker cone and a guitar string.

## 43. SLO P-10-D-13 - Describe the longitudinal nature of sound.

Answer

Sound in air is longitudinal. Air particles vibrate back and forth parallel to the direction of sound propagation.

This produces alternating compressions and rarefactions. Energy travels through the air, but particles mainly oscillate about equilibrium positions.

## 44. SLO P-10-D-14 - State the approximate audible frequency range for humans.

Answer

For a young person with healthy hearing:

> Audible range ≈ 20 Hz to 20,000 Hz

Below 20 Hz is infrasound; above 20,000 Hz is ultrasound. The upper hearing limit generally decreases with age.

## 45. SLO P-10-D-15 - Explain why sound cannot travel in a vacuum and describe an experiment.

Answer

Sound is mechanical and requires particles to transfer the disturbance. A vacuum has no material medium, so sound cannot propagate through it.

**Bell-jar experiment:** Place a ringing electric bell in a sealed bell jar. Pump out the air. The hammer remains visibly vibrating, but the sound becomes progressively fainter. When air is readmitted, the sound becomes louder.

**Conclusion:** Sound requires a material medium.

## 46. SLO P-10-D-16 - Explain how amplitude and frequency affect loudness and pitch.

Answer

Increasing amplitude increases sound intensity and generally makes sound louder. Decreasing amplitude makes it softer. Increasing frequency produces a higher pitch; decreasing it produces a lower pitch.

For otherwise comparable waves:

$$\text{Intensity} \propto A^2$$

On an oscilloscope, louder sound has greater vertical amplitude, while higher-pitched sound has more oscillations in the same time interval.

## 47. SLO P-10-D-17 - Explain timbre using sound sources and oscilloscope traces.

Answer

Timbre, or quality, allows us to distinguish sounds of the same pitch and loudness from different sources.

Different instruments produce different mixtures and strengths of harmonics, giving different waveform shapes.

On an oscilloscope, two sounds may have the same period and similar amplitude but different trace shapes. A flute and violin playing the same note sound different because their harmonic contents differ.

## 48. SLO P-10-D-18 - Describe an echo as reflected sound.

Answer

An echo is reflected sound heard separately from the original. A distinct echo generally requires a delay of about 0.1 s or more.

For a source and listener close together:

$$d = vt / 2$$

At $v = 340\ \text{m s}^{-1}$:

$$d(\text{minimum}) = (340 \times 0.1) / 2 = 17\ \text{m}$$

## 49. SLO P-10-D-19 - Describe a simple experiment showing reflection of sound.

Answer

Place two tubes facing a large, hard reflecting board. Put a ticking clock or sound source at one tube's outer end and listen through the other tube. Adjust it until the sound is loudest.

The loudest sound occurs when the tubes make approximately equal angles with the normal.

**Conclusion:** Sound reflects from a hard surface, with angle of incidence equal to angle of reflection.

## 50. SLO P-10-D-20 - Describe a distance-time method for determining the speed of sound.

Answer

**Echo method:** Stand a measured distance d from a large wall. Produce a sharp sound and measure time t until its echo. Repeat and calculate the mean time.

Sound travels to the wall and back:

$$v = 2d / t$$

**Example:** d = 85 m, t = 0.50 s.

$$v = (2 \times 85) / 0.50 = 340 \text{ m s}^{-1}$$

Use a sufficiently large distance or electronic timing to reduce timing error.

## 51. SLO P-10-D-21 - State the approximate speed of sound in air.

Answer

The speed of sound in air is approximately 330-350 m $s^{-1}$. A common school-level value is:

$$v = 340 \text{ m s}^{-1}$$

Its value increases with air temperature.

## 52. SLO P-10-D-22 - Compare the speed of sound in solids, liquids and gases.

Answer

In general:

$$v(\text{solids}) > v(\text{liquids}) > v(\text{gases})$$

Solids usually transmit disturbances rapidly because particles interact through strong elastic forces.

Typical approximate values: steel 5000 m $s^{-1}$; water 1500 m $s^{-1}$; air 340 m $s^{-1}$.

The exact speed depends on elastic properties and density.

## 53. SLO P-10-D-23 - Define ultrasound.

Answer

Ultrasound is sound with frequency greater than the upper limit of normal human hearing:

$$f > 20{,}000 \text{ Hz} = 20 \text{ kHz}$$

It is a mechanical wave and requires a medium.

## 54. SLO P-10-D-24 - Explain ultrasound in cleaning, medical scanning and sonar, including calculations.

Answer

**Cleaning:** Rapid pressure changes in liquid produce tiny bubbles whose formation and collapse help remove dirt from difficult-to-reach surfaces.

**Medical scanning:** Ultrasound pulses reflect at boundaries between tissues. A computer uses returning echoes to construct an image. Uses include prenatal scanning and examination of internal organs.

**Sonar:** Pulses reflect from the seabed or an underwater object. Distance is:

$$d = vt / 2$$

**Example:** Echo time = 0.40 s; sound speed in water = 1500 m $s^{-1}$.

$$d = (1500 \times 0.40) / 2 = 300 \text{ m}$$

## 55. SLO P-10-D-25 - Explain uses of infrasound.

Answer

Infrasound has frequency below 20 Hz.

- **Animal communication:** Elephants use low-frequency sounds, including infrasound, over long distances.
- **Geophysical monitoring:** Infrasonic signals help monitor volcanic eruptions, explosions and atmospheric effects associated with major seismic events.

Long wavelengths allow strong diffraction, and low-frequency sound can travel considerable distances under suitable conditions.

## 56. SLO P-10-D-26 - Analyze the environmental effects of noise pollution.

Answer

Noise pollution is unwanted or excessive sound disturbing organisms and normal activities.

Effects include hearing damage from prolonged high-level exposure, stress, disturbed sleep, reduced concentration, interference with speech, disruption of animal feeding and breeding, and disturbance of marine animals by underwater noise.

Traffic, machinery, aircraft and loudspeakers are common sources.

## 57. SLO P-10-D-27 - Justify the importance of acoustic protection.

Answer

Acoustic protection reduces harmful or disturbing sound exposure. It protects hearing, improves communication and concentration, and reduces disturbance.

Methods include earplugs or earmuffs, enclosures around noisy machines, roadside sound barriers, absorbing materials in buildings, reduced exposure time and control at the noise source.

## 58. SLO P-10-D-28 - Explain how sound properties are applied in building acoustics.

Answer

Good acoustic design provides clear, evenly distributed sound without excessive echoes or reverberation.

Buildings use absorbing wall and ceiling materials, suitable room shapes to avoid sound focusing, reflectors to direct useful sound, insulation against outside noise, and careful loudspeaker placement.

Suitable reverberation time depends on whether the room is mainly for speech or music.

## 59. SLO P-10-D-29 - Explain how soft materials reduce echoes.

Answer

Soft, porous materials absorb sound energy. Sound entering pores makes air and fibres vibrate. Some energy becomes thermal energy through friction and other losses, so less is reflected.

Curtains, carpets, upholstered seats and acoustic panels reduce echoes and excessive reverberation in classrooms, halls and studios.

## 60. SLO P-10-D-30 - Explain reflection, refraction and diffraction of sound with examples.

Answer

- **Reflection:** Sound returns from a boundary, as in an echo from a wall.
- **Refraction:** Sound changes direction because its speed changes, as through air layers at different temperatures.
- **Diffraction:** Sound spreads around obstacles or through openings, allowing a voice to be heard around a doorway.

Lower-frequency sound generally diffracts more strongly around the same obstacle.

# 5. Geometrical optics

## 61. SLO P-10-D-32 - Define normal, angle of incidence and angle of reflection.

Answer

- **Normal:** A line perpendicular to the reflecting surface at the point where the ray strikes it.
- **Angle of incidence, i:** Angle between the incident ray and normal.
- **Angle of reflection, r:** Angle between the reflected ray and normal.

Both angles are measured from the normal, not from the surface.

## 62. SLO P-10-D-33 - Describe an experiment to locate and identify a plane-mirror image.

Answer

1. Place a vertical glass plate on a sheet of paper.

2. Put an object pin in front of it and a second pin behind it.

3. Move the second pin until it coincides with the first pin's image, with no relative movement when the eye moves sideways. This is the no-parallax position.

4. Measure the perpendicular distances of the two pins from the plate.

**Observations:** The image is the same distance behind the plate as the object is in front and is the same size.

**Characteristics:** Virtual, upright, laterally inverted and equal in size to the object.

## 63. SLO P-10-D-34 - Apply the laws of reflection.

Answer

1. The incident ray, reflected ray and normal lie in the same plane.

2. Angle of incidence equals angle of reflection:

```
i = r
```

**Example:** A ray makes 30° with the mirror surface.

```
i = 90° - 30° = 60°; therefore r = 60°
```

## 64. SLO P-10-D-35 - Define normal, angle of incidence and angle of refraction.

Answer

- **Normal:** A line perpendicular to the boundary at the point of incidence.
- **Angle of incidence, i:** Angle between the incident ray and normal.
- **Angle of refraction, r:** Angle between the refracted ray and normal in the second medium.

Angles are measured from the normal.

## 65. SLO P-10-D-36 - Explain bending towards or away from the normal during refraction.

Answer

When a wave enters a medium where it slows down, it bends towards the normal. When it speeds up, it bends away from the normal.

At normal incidence it continues along the normal without bending, although speed and wavelength may change.

Frequency stays constant during refraction:

$$v = f\lambda$$

Therefore, decreasing speed decreases wavelength.

## 66. SLO P-10-D-37 - Define and use refractive index n = c/v.

Answer

Absolute refractive index is the ratio of light speed in vacuum to its speed in the medium:

$$n = c / v$$

It has no unit.

**Example:** $v = 2.0 \times 10^8$ m $s^{-1}$.

$$n = (3.0 \times 10^8) / (2.0 \times 10^8) = 1.5$$

## 67. SLO P-10-D-38 - Define refractive index using angles and apply Snell's law.

Answer

For air to a medium:

$$n \approx \sin i / \sin r$$

General Snell's law:

$$n_1 \sin i = n_2 \sin r$$

**Example:** A ray enters glass of index 1.5 from air at i = 30°.

$$1 \times \sin 30° = 1.5 \sin r$$

$$\sin r = 0.5 / 1.5 = 1/3; \text{ therefore } r \approx 19.5°$$

## 68. SLO P-10-D-40 - Define critical angle and total internal reflection.

Answer

The critical angle C is the incidence angle in the optically denser medium for which the refraction angle in the rarer medium is 90°.

Total internal reflection is complete reflection of light back into the denser medium.

**Conditions:** Light must travel from a higher to a lower refractive index, and incidence angle must exceed the critical angle:

$$i > C$$

## 69. SLO P-10-D-41 - Derive n = 1/sin C.

Answer

At a medium-air boundary, Snell's law gives:

$$n \sin i = n(air) \sin r$$

At the critical angle, i = C, r = 90° and n(air) ≈ 1.

$$n \sin C = 1 \times \sin 90° = 1$$

Hence:

$$n = 1 / \sin C$$

This form applies when the second medium is air or approximately vacuum.

## 70. SLO P-10-D-42 - Use n = 1/sin C in calculations.

Answer

$$n = 1 / \sin C \text{ and } C = \sin^{-1}(1/n)$$

**Example:** For glass with n = 1.5:

$$C = \sin^{-1}(1/1.5) \approx 41.8°$$

For glass to air, total internal reflection occurs for i > 41.8°.

## 71. SLO P-10-D-43 - Describe experiments showing internal reflection and total internal reflection.

Answer

1. Place a semicircular glass block on paper.

2. Direct a narrow ray through the curved surface towards the centre, so it enters approximately normally.

3. Let it strike the flat glass-air boundary from inside.

4. Increase the internal incidence angle gradually.

**Observations:** Below C, both reflected and refracted rays appear. At C, the refracted ray runs along the boundary. Above C, no refracted ray emerges; total internal reflection occurs.

## 72. SLO P-10-D-44 - Explain optical fibres and their advantages in telecommunications.

Answer

An optical fibre has a transparent core surrounded by cladding of lower refractive index. Suitably entering light travels along the core by repeated total internal reflection at the core-cladding boundary.

Information is encoded into light pulses.

**Advantages:** High information capacity; low signal loss over long distances; immunity to electromagnetic interference; small size and low mass; electrical insulation; and greater difficulty of intercepting signals without detection.

## 73. SLO P-10-D-45 - Explain how converging and diverging lenses affect parallel light.

Answer

A converging lens brings rays parallel to its principal axis together at the principal focus on the opposite side.

A diverging lens spreads parallel rays apart. Their backward extensions meet at the principal focus on the incident side.

For ordinary glass lenses in air, a convex lens is converging and a concave lens is diverging.

## 74. SLO P-10-D-46 - Define focal length, principal axis and principal focus.

Answer

- **Principal axis:** Straight line through the optical centre and centres of curvature of the lens surfaces.
- **Converging-lens principal focus:** Point where rays initially parallel to the principal axis converge.
- **Diverging-lens principal focus:** Point from which initially parallel rays appear to diverge.
- **Focal length:** Distance between optical centre and principal focus.

## 75. SLO P-10-D-47 - Draw ray diagrams showing real and virtual images formed by a converging lens.

Answer

Use two principal rays from the object's tip: (1) a ray parallel to the axis refracts through the far focus; (2) a ray through the optical centre continues approximately undeviated. Their intersection, or that of backward extensions, locates the image.

| Object position | Image position | Characteristics |
|---|---|---|
| Beyond 2F | Between F and 2F, opposite side | Real, inverted, diminished |
| At 2F | At 2F, opposite side | Real, inverted, same size |
| Between F and 2F | Beyond 2F, opposite side | Real, inverted, magnified |
| At F | At infinity in the ideal model | Parallel emerging rays; no finite image |
| Between F and lens | Same side as object | Virtual, upright, magnified |

**Diagram:** See Appendix A for representative real- and virtual-image ray diagrams.

## 76. SLO P-10-D-48 - Differentiate between real and virtual images.

Answer

| Real image | Virtual image |
|---|---|
| Actual convergence of rays | Apparent meeting of backward ray extensions |
| Can be formed on a screen | Cannot be formed directly on a screen |
| For a single converging lens with a real object, usually inverted | For a converging lens used as a magnifier, upright |
| Example: image on a camera sensor | Example: plane-mirror image |

## 77. SLO P-10-D-49 - Define and calculate linear magnification.

Answer

Linear magnification is image height divided by object height:

$$m = h_i / h_o$$

For size comparisons use magnitudes: $|m| = |h_i| / |h_o|$.

**Example:** Object height = 2 cm; image height = 6 cm.

$$|m| = 6 / 2 = 3$$

The image is three times the object's size. With signed heights, negative magnification indicates an inverted image.

## 78. SLO P-10-D-50 - Describe a single lens used as a magnifying glass.

Answer

A magnifying glass is a converging lens with its object between the lens and principal focus:

$$d_o < f$$

The emerging rays diverge, but their backward extensions meet on the object's side. The image is virtual, upright and magnified.

**Diagram:** See Appendix A, virtual-image diagram.

## 79. SLO P-10-D-51 - Explain prism dispersion and detection of non-visible radiation using a thermometer.

Answer

Dispersion is the separation of white light into component colours. A prism refracts different wavelengths by different amounts because its refractive index depends on wavelength. In ordinary glass, violet deviates more than red.

**Infrared experiment:** Produce a spectrum using sunlight and a prism. Put thermometers in different regions, including beyond the red end, and use a shaded control thermometer. A

temperature rise beyond red shows invisible infrared radiation.

A thermometer demonstrates infrared; ultraviolet is more suitably detected by fluorescence or an appropriate sensor.

## 80. SLO P-10-D-52 - State the seven visible colours in frequency and wavelength order.

Answer

**Increasing frequency:** Red → Orange → Yellow → Green → Blue → Indigo → Violet.

**Increasing wavelength:** Violet → Indigo → Blue → Green → Yellow → Orange → Red.

Red has the longest wavelength and lowest frequency of these colours; violet has the shortest wavelength and highest frequency.

## 81. SLO P-10-D-54 - Draw image-formation diagrams for normal, short-sighted and long-sighted eyes.

Answer

| Eye | Image formation |
|---|---|
| Normal | Light focuses on the retina. |
| Short-sighted: myopia | Light from distant objects focuses in front of the retina. |
| Long-sighted: hyperopia | Without sufficient accommodation, light would focus behind the retina, especially for near objects. |

A myopic eye may be too long or have excessive converging power. A hyperopic eye may be too short or have insufficient converging power.

**Diagram:** See Appendix A. Label the eye lens, principal axis, rays, focus and retina.

## 82. SLO P-10-D-55 - Explain lenses used to correct short- and long-sightedness.

Answer

| Defect | Correcting lens | Action |
|---|---|---|
| Short-sightedness | Concave/diverging | Diverges incoming light so the eye focuses it farther back, on the retina. |
| Long-sightedness | Convex/converging | Adds converging power so the eye focuses it farther forward, on the retina. |

**Diagram:** See the corrected-eye diagrams in Appendix A.

# 6. Electrostatics

## 83. SLO P-10-E-01 - State the two types of charge and their unit.

Answer

There are positive and negative electric charges. The SI unit is the coulomb (C).

A proton has charge $+1.6 \times 10^{-19}$ C; an electron has charge $-1.6 \times 10^{-19}$ C.

A neutral object has equal total positive and negative charge.

## 84. SLO P-10-E-02 - State the interaction of like and unlike charges.

Answer

Like charges repel; unlike charges attract. Therefore, two positive charges repel, two negative charges repel, and a positive and negative charge attract.

For point charges, the forces act along the line joining them.

## 85. SLO P-10-E-03 - Describe experiments showing charging by friction.

Answer

Rub a plastic rod with a dry cloth and bring it near small paper pieces. Their attraction shows that the rod has become charged.

Suspend one of two similarly rubbed plastic rods and bring the other near it. Repulsion shows similar charges.

A glass rod rubbed with silk usually becomes positive while silk becomes negative. Contact and separation transfer electrons between the materials.

## 86. SLO P-10-E-04 - Explain that frictional charging of solids involves electron transfer.

Answer

Electrons transfer from one solid to another. An object gaining electrons becomes negative; one losing electrons becomes positive.

Protons remain bound in atomic nuclei and are not transferred between solids. Charge is conserved: the charge gained by one equals that lost by the other.

## 87. SLO P-09-E-05 - Explain discharge of an insulator above a flame and in damp conditions.

Answer

**Code note:** The document prints this SLO as P-09-E-05.

**Flame:** A flame produces ions in surrounding air. Oppositely charged ions move to the insulator and neutralize its surface charge.

**Damp conditions:** Moisture forms a slightly conducting surface layer, allowing charge to leak to the surroundings or earth.

A dry insulator does not discharge easily because charge cannot move freely through it.

## 88. SLO P-10-E-06 - Explain charging a conductor by induction and earthing.

Answer

To charge a metal sphere positively using a negative rod:

1. Place the sphere on an insulating stand.
2. Bring the negative rod near without touching. Electrons move away from it.
3. Earth the far side with the rod still nearby. Electrons flow to Earth.
4. Remove the earth connection first, then remove the rod.

The sphere remains positively charged, with charge distributed over its surface. The permanent induced charge is opposite to that of the inducing rod.

## 89. SLO P-10-E-07 - Describe problems caused by electrostatic charging.

Answer

Electrostatic charge can produce strong electric fields and sparks.

- **Lightning:** May damage buildings and injure people.
- **Fuel handling:** Sparks may ignite flammable vapour.
- **Electronics:** Electrostatic discharge may damage sensitive circuits.
- **Dust:** Charged surfaces attract dust and may interfere with equipment.
- **Clothing:** Static causes clinging and small shocks.

## 90. SLO P-10-E-08 - Explain charging and discharging in photocopiers and electrostatic precipitators.

Answer

**Photocopier:** A photoconductive drum is charged. Reflected light discharges selected regions, forming an electrostatic image. Charged toner is attracted to appropriate regions, transferred to paper, then fixed by heat and pressure.

**Electrostatic precipitator:** Dust or smoke particles receive charge in a strong field and move to oppositely charged or suitably earthed collecting plates. The particles are removed from the plates, reducing industrial particulate pollution.

## 91. SLO P-10-E-09 - Define an electric field.

Answer

An electric field is a region where an electric charge experiences an electric force.

Electric field strength is force per unit positive test charge:

$$E = F / q$$

Its SI unit is N $C^{-1}$. Electric field strength is a vector quantity.

## 92. SLO P-10-E-10 - State the direction of an electric field line.

Answer

The field direction at a point is the force direction on a small positive test charge there.

Field lines point away from positive charges and towards negative charges. A tangent to a field line gives the field direction at that point.

## 93. SLO P-10-E-11 - Describe electric field patterns around charges, conducting spheres and parallel plates.

Answer

| Arrangement | Field pattern |
|---|---|
| Positive point charge | Radial lines outwards |
| Negative point charge | Radial lines inwards |
| Isolated charged conducting sphere | Outside: radial lines perpendicular to its surface, with direction set by charge sign |
| Oppositely charged parallel plates | Straight, parallel, equally spaced lines from positive to negative, ignoring edges |

The field between plates is approximately uniform. Inside a conductor in electrostatic equilibrium, the field is zero.

**Drawing rules:** Add arrowheads; never cross field lines; show closer spacing for a stronger field.

## 94. SLO P-10-E-12 - Give examples of electrical conductors and insulators.

Answer

| Conductors | Insulators |
|---|---|
| Copper | Rubber |
| Aluminium | Glass |
| Silver | Dry plastic |
| Graphite | Dry wood |
| Salt solution | Air under ordinary conditions |

Conductors allow charge to move relatively easily; insulators strongly resist its movement.

## 95. SLO P-10-E-13 - Describe an experiment distinguishing conductors from insulators.

Answer

Connect a cell, switch and small lamp in series, leaving a gap. Place the test material across the gap and close the switch.

If the lamp lights, the material conducts sufficiently to complete the circuit. If not, the material may be an insulator or have very high resistance.

A sensitive current detector is more reliable for weak conductors. Keep the supply and connections unchanged when comparing materials.

## 96. SLO P-10-E-14 - Use the electron model to distinguish conductors and insulators.

Answer

Metals contain free or delocalized electrons able to move through the material. In an applied electric field they acquire a net drift, producing current.

In insulators, electrons are tightly bound and cannot move freely through the material. Metals therefore conduct well, whereas insulators conduct poorly under ordinary conditions.

## 97. SLO P-10-E-15 - Explain how a lightning rod protects humans.

Answer

A lightning protection system has a metal air terminal connected by a thick conductor to an effective earth connection.

If lightning strikes, the conductor provides a low-resistance path for current to Earth. This reduces current through the building or people, and reduces fire and structural damage.

The system must be continuous and properly earthed.

## 98. SLO P-10-E-16 - Explain electrical breakdown of a gas.

Answer

Normally, a gas conducts poorly. In a sufficiently strong field, free electrons accelerate and collide with atoms or molecules, releasing more electrons.

An avalanche of charged particles develops, ionizing the gas and making it conducting. A large current can then pass through the ionized path, producing a spark or discharge.

## 99. SLO P-10-E-17 - Describe corona discharge and Lichtenberg figures as examples of breakdown.

Answer

**Corona discharge:** Localized ionization around a highly charged conductor, especially near sharp points, sometimes producing a faint glow.

**Lichtenberg figures:** Branching patterns produced by discharge over or through insulating materials.

Both show effects of strong electric fields and electrical breakdown.

## 100. SLO P-10-E-18 - Explain the formation of lightning.

Answer

1. Interactions among ice particles, droplets and other particles separate charge within thunderclouds. Similar charging occurs among volcanic ash particles.

2. Oppositely charged regions create strong electric fields.

3. A charged cloud induces opposite surface charge on the ground beneath it.

4. A sufficiently strong field causes air breakdown.

5. Conducting channels, including leaders and streamers, develop.

6. When a connection forms, a large discharge creates the flash.

Rapid air heating and expansion produce thunder.

# 7. Current electricity and circuits

## 101. SLO P-10-E-20 - Define and calculate electric current.

Answer

Current is the rate of charge flow through a cross-section:

$$I = Q / t$$

Its SI unit is the ampere.

**Example:** 12 C passes in 4 s:

$$I = 12 / 4 = 3 \text{ A}$$

For electron-number calculations, Q = Ne, so:

$$N = It / e$$

## 102. SLO P-10-E-21 - Explain electrical conduction in metals using free electrons.

Answer

Metals contain free electrons moving randomly among positive ions. Without an applied potential difference, random motion gives no net current.

An applied potential difference establishes an electric field. Electrons acquire a small average drift opposite to the field, producing current.

Collisions with the lattice oppose drift and contribute to resistance.

## 103. SLO P-10-E-22 - State the unit of current and its relationship to charge and time.

Answer

Current is measured in amperes (A). One ampere means one coulomb passing a point each second:

$$1 \text{ A} = 1 \text{ C s}^{-1}$$

## 104. SLO P-10-E-23 - Differentiate between direct and alternating current.

Answer

| Direct current: DC | Alternating current: AC |
|---|---|
| Flows in one direction | Reverses direction periodically |
| May be steady or vary in magnitude | Usually varies periodically in magnitude |
| Cells and batteries supply DC | AC generators and mains supply AC |
| Steady-DC current-time graph is horizontal | Sinusoidal-AC graph crosses above and below zero |

For sinusoidal AC:

$$I = I_0 \sin(2\pi ft)$$

Negative current means flow opposite to the chosen positive direction.

## 105. SLO P-10-E-24 - Differentiate conventional and actual current.

Answer

**Conventional current:** Direction in which positive charge would move: positive to negative through an external circuit.

**Electron flow in metals:** Electrons move from negative to positive through the external circuit, opposite to conventional current.

Here 'actual current' refers to electron flow in metals; other media may have both positive and negative charge carriers.

## 106. SLO P-10-E-26 - Define and calculate electromotive force.

Answer

Emf is energy supplied by a source per unit charge passing through it:

$$\mathcal{E} = W / Q$$

It is measured in volts.

**Example:** A source supplies 24 J to 4 C:

$$\mathcal{E} = 24 / 4 = 6 \text{ V}$$

Despite its name, emf is not a mechanical force.

## 107. SLO P-10-E-27 - Define and calculate potential difference.

Answer

Potential difference across a component is energy transferred per unit charge passing through it:

$$V = W / Q$$

**Example:** A lamp transfers 60 J when 5 C passes:

$$V = 60 / 5 = 12 \text{ V}$$

Emf describes energy supplied per coulomb by a source; potential difference describes energy transferred per coulomb between two points.

## 108. SLO P-10-E-28 - State the unit of emf and potential difference.

Answer

Both are measured in volts (V). One volt means one joule transferred per coulomb:

$$1 \text{ V} = 1 \text{ J C}^{-1}$$

## 109. SLO P-10-E-30 - Calculate the total emf of sources connected in series.

Answer

For series sources in the same aiding direction:

$$\mathcal{E}(\text{total}) = \mathcal{E}_1 + \mathcal{E}_2 + \ldots$$

**Example:** Three 1.5 V cells give:

$$\mathcal{E}(\text{total}) = 1.5 + 1.5 + 1.5 = 4.5 \text{ V}$$

For opposing sources, subtract the opposing emf. Example: 6 V - 1.5 V = 4.5 V.

## 110. SLO P-10-E-31 - State the emf of identical sources connected in parallel.

Answer

Identical sources in parallel, with like terminals together, have the same emf as one source.

**Example:** Three identical 1.5 V cells in parallel give:

$$\mathcal{E}(\text{total}) = 1.5 \text{ V}$$

This increases available capacity and can share load current; the emfs do not add.

## 111. SLO P-10-E-33 - Define resistivity and use resistance-length-area relationships.

Answer

Resistivity is a material property defined by:

$$\rho = RA / L \text{ or } R = \rho L / A$$

Its SI unit is Ω m. For the same material at constant temperature:

$$R \propto L \text{ and } R \propto 1/A$$

**Example:** R = 2 Ω, A = $1.0 \times 10^{-6}$ m², L = 4 m.

$$\rho = (2 \times 1.0 \times 10^{-6}) / 4 = 5.0 \times 10^{-7} \ \Omega \text{ m}$$

## 112. SLO P-10-E-34 - State and apply Ohm's law at constant temperature.

Answer

Current through an ohmic conductor is directly proportional to potential difference, provided temperature and other physical conditions remain constant.

$$V \propto I \text{ therefore } V = IR$$

**Example:** V = 12 V and I = 0.50 A.

$$R = V / I = 12 / 0.50 = 24\ \Omega$$

A constant-resistance conductor has a straight-line current-voltage graph through the origin.

## 113. SLO P-10-E-35 - Explain the effect of temperature on a metallic resistor or lamp filament.

Answer

Heating a metal increases lattice-ion vibration. Free electrons experience more scattering, so resistance usually increases.

In a lamp, increasing voltage increases current and filament temperature. The hotter filament has greater resistance, so current increases less rapidly than voltage.

The lamp is therefore non-ohmic across a range where its temperature changes.

## 114. SLO P-10-E-36 - Interpret current-voltage graphs for a resistor, filament lamp and diode.

Answer

| Component | Graph and interpretation |
|---|---|
| Constant-resistance resistor | Straight line through origin: $I \propto V$. |
| Filament lamp | Curve becomes less steep as voltage magnitude rises because heating increases resistance. |
| Diode | Very small reverse current before breakdown; forward current rises rapidly after sufficient forward voltage. |

For an operating point:

$$R = V / I$$

For a straight-line I-against-V graph, gradient = 1/R.

**Diagram:** See Appendix B. Current is vertical; voltage is horizontal.

## 115. SLO P-10-E-37 - Draw circuit diagrams using the listed circuit components.

Answer

Use standard symbols, shown in Appendix C.

| Component | What to show |
|---|---|
| Cell | One long and one short parallel line; long line positive |
| Battery | More than one cell symbol |
| Power supply | Clearly marked supply terminals or standard symbol |
| Generator | Generator symbol; indicate AC where appropriate |
| Fixed resistor | Standard resistor symbol |
| Variable resistor | Resistor with a diagonal arrow |
| Potential divider | Resistor track with movable contact |
| Switch | Open or closed contacts |
| Heater | Heating-resistor symbol |
| NTC thermistor | Thermistor symbol |
| LDR | Resistor with arrows pointing towards it |
| Lamp | Circle containing a cross |
| Motor | Circle containing M |
| Ammeter | Circle containing A; in series |
| Voltmeter | Circle containing V; in parallel |
| Transformer | Two magnetically coupled coils |
| Fuse | Standard fuse symbol |
| Relay | Coil linked to a separate switch |
| Diode | Diode symbol with cathode bar |
| LED | Diode symbol with outward arrows |

Draw neat wires and clear junctions. Place an ammeter in series and a voltmeter across the component being measured.

## 116. SLO P-10-E-38 - Apply current and voltage distribution rules in series and parallel circuits.

Answer

**Series:** Current is the same throughout; supply voltage equals the sum of individual voltage drops.

$I = I_1 = I_2 = \ldots$ and $V(\text{supply}) = V_1 + V_2 + \ldots$

**Parallel:** Total current equals the sum of branch currents; each branch has the same potential difference.

$$I(total) = I_1 + I_2 + \ldots \text{ and } V(supply) = V_1 = V_2 = \ldots$$

Current conservation follows charge conservation; the voltage rule follows energy conservation.

**Example:** If 5 A enters a junction and one outgoing branch carries 2 A, the other carries 5 - 2 = 3 A.

## 117. SLO P-10-E-39 - Calculate equivalent resistance in series.

Answer

Series resistances add:

$$R(eq) = R_1 + R_2 + R_3 + \ldots$$

**Example:** $R_1 = 2\ \Omega$, $R_2 = 3\ \Omega$ and $R_3 = 5\ \Omega$.

$$R(eq) = 2 + 3 + 5 = 10\ \Omega$$

Equivalent series resistance exceeds any individual resistance.

## 118. SLO P-10-E-40 - Calculate equivalent resistance of two parallel resistors.

Answer

$$1/R(eq) = 1/R_1 + 1/R_2$$

For two resistors:

$$R(eq) = R_1R_2 / (R_1 + R_2)$$

**Example:** 6 Ω and 3 Ω in parallel:

$$R(eq) = (6 \times 3) / (6 + 3) = 2\ \Omega$$

The equivalent resistance is smaller than the smallest branch resistance.

## 119. SLO P-10-E-41 - Calculate current, voltage and resistance in whole circuits and individual parts.

Answer

Simplify the resistances, calculate total current, then apply series and parallel rules.

**Example:** A 2 Ω resistor is in series with two parallel 6 Ω resistors, across 10 V.

$$R(parallel) = (6 \times 6) / (6 + 6) = 3\ \Omega$$

$$R(total) = 2 + 3 = 5\ \Omega$$

$$I(total) = 10 / 5 = 2\ A$$

Voltage across the 2 Ω resistor:

$$V = IR = 2 \times 2 = 4\ V$$

Voltage across the parallel section:

$$V(\text{parallel}) = 10 - 4 = 6\ \text{V}$$

Current in each 6 Ω branch:

$$I(\text{branch}) = 6 / 6 = 1\ \text{A}$$

## 120. SLO P-10-E-42 - Explain NTC thermistors and LDRs as input sensors.

Answer

| Sensor | Response | Use |
|---|---|---|
| NTC thermistor | Resistance decreases as temperature rises | Temperature control, alarms and thermostats |
| LDR | Resistance decreases as light intensity rises | Automatic lighting and light detectors |

They are often used in potential dividers. Their changing resistance changes the output voltage, allowing an electronic circuit to respond to temperature or light.

## 121. SLO P-10-E-43 - Explain variable potential dividers and use voltage-resistance ratios.

Answer

A potential divider uses series resistors to provide a fraction of the supply voltage.

$$V_1 / V_2 = R_1 / R_2$$

For an unloaded output across $R_2$:

$$V(\text{out}) = V(\text{supply}) \times R_2 / (R_1 + R_2)$$

**Example:** $V(\text{supply}) = 12$ V, $R_1 = 2$ kΩ, $R_2 = 4$ kΩ.

$$V(\text{out}) = 12 \times 4 / (2 + 4) = 8\ \text{V}$$

Moving a sliding contact or changing a sensor resistance varies the output voltage.

## 122. SLO P-10-E-44 - Explain and use resistor colour codes.

Answer

Bands identify resistance and tolerance on resistors too small for numerical labels.

For four bands: first = first digit; second = second digit; third = multiplier; fourth = tolerance.

| Colour | Digit | Multiplier |
|---|---|---|
| Black | 0 | $10^0$ |
| Brown | 1 | $10^1$ |
| Red | 2 | $10^2$ |
| Orange | 3 | $10^3$ |

| Colour | Digit | Multiplier |
|---|---|---|
| Yellow | 4 | $10^4$ |
| Green | 5 | $10^5$ |
| Blue | 6 | $10^6$ |
| Violet | 7 | $10^7$ |
| Grey | 8 | $10^8$ |
| White | 9 | $10^9$ |

Gold multiplier = $10^{-1}$; silver = $10^{-2}$. Common tolerances: gold ±5%, silver ±10%, brown ±1%, red ±2%.

**Example:** Brown-black-red-gold:

$R = 10 \times 10^2 = 1000\ \Omega = 1.0\ k\Omega$, ±5%

# 8. Electrical appliances and mains

## 123. SLO P-10-E-48 - State common uses of electricity.

Answer

| Use | Example | Main energy conversion |
|---|---|---|
| Heating | Electric heater | Electrical → thermal |
| Lighting | LED lamp | Electrical → light, with some heating |
| Battery charging | Battery charger | Electrical → chemical energy stored |
| Motors | Fan or pump | Electrical → mechanical |
| Electronic systems | Computer or communication equipment | Electrical energy powers processing and transmission |

## 124. SLO P-10-E-49 - Explain the advantages of connecting lamps in parallel.

Answer

1. Each lamp receives the full supply voltage.

2. Each can be switched independently.

3. Failure of one does not interrupt the others.

4. Different power ratings can operate at their rated voltage.

5. Adding a branch does not significantly change voltage across existing lamps when the supply is suitable.

These advantages make parallel connections appropriate for household lighting.

## 125. SLO P-10-E-50 - Apply P = IV.

Answer

Electrical power is the rate of electrical energy transfer:

$$P = IV$$

Its SI unit is the watt.

**Example:** An appliance operates at 230 V and draws 2 A:

$$P = 230 \times 2 = 460\ W$$

For an ohmic resistor:

$$P = I^2R = V^2/R$$

## 126. SLO P-10-E-51 - Apply E = IVt.

Answer

For constant voltage and current:

$$E = IVt = Pt$$

Use seconds to obtain joules.

**Example:** A 12 V device draws 2 A for 5 minutes:

$$t = 5 \times 60 = 300 \text{ s}$$

$$E = 2 \times 12 \times 300 = 7200 \text{ J}$$

## 127. SLO P-10-E-52 - Define the kilowatt-hour.

Answer

One kilowatt-hour is energy transferred by an appliance operating at 1 kW for 1 hour.

$$1 \text{ kWh} = 1000 \text{ W} \times 3600 \text{ s} = 3.6 \times 10^6 \text{ J}$$

A kilowatt-hour is a unit of energy, not power. One electricity billing 'unit' normally means 1 kWh.

## 128. SLO P-10-E-53 - Explain why components need suitable power ratings.

Answer

A component's power rating indicates power it can safely dissipate or handle under specified conditions. Exceeding it may cause overheating, damage or failure.

Calculate using $P = VI$, $P = I^2R$ or $P = V^2/R$.

**Example:** A 100 Ω resistor across 10 V dissipates:

$$P = 10^2 / 100 = 1 \text{ W}$$

A 0.25 W resistor is unsuitable. A rating above calculated dissipation provides an operating margin.

## 129. SLO P-10-E-54 - Calculate electricity costs using kWh.

Answer

$$E(\text{kWh}) = P(\text{kW}) \times t(\text{hours})$$

$$\text{Cost} = E(\text{kWh}) \times \text{price per kWh}$$

**Example:** Four 100 W bulbs operate 6 hours daily for 30 days at Rs. 12 per kWh.

$$P = 4 \times 100 = 400 \text{ W} = 0.4 \text{ kW}$$

$$t = 6 \times 30 = 180 \text{ hours}$$

$$E = 0.4 \times 180 = 72 \text{ kWh}$$

$$\text{Cost} = 72 \times 12 = \text{Rs. } 864$$

This excludes any additional billing charges.

## 130. SLO P-10-E-57 - Explain what happens when a live wire touches an earthed metal case.

Answer

1. The live wire creates a fault path through the metal case and earth conductor.

2. The low-resistance path allows a large fault current.

3. The protective fuse or circuit breaker disconnects the supply.

4. Earthing and rapid disconnection reduce electric-shock risk.

The earth connection should have low resistance and remain securely attached to the case.

## 131. SLO P-10-E-58 - Explain why an appliance's outer casing must be non-conducting or earthed.

Answer

**Earthed metal casing:** A live-wire fault creates a low-resistance path through earth, allowing the protective device to disconnect the supply.

**Double-insulated appliance:** Insulating barriers separate live parts from accessible surfaces. The non-conducting outer casing cannot normally become live.

Both arrangements reduce electric-shock risk. Properly double-insulated appliances do not need an earth connection.

## 132. SLO P-10-E-59 - State the three wires in a mains circuit and their functions.

Answer

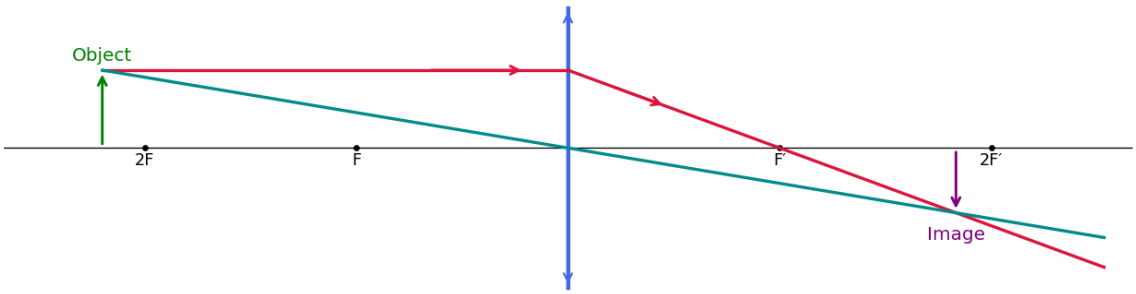
| Wire | Function |
|---|---|
| Live or line | Alternating potential relative to earth; supplies energy to the appliance |
| Neutral | Normal return path; maintained near earth potential at the supply system |
| Earth | Connects exposed conducting parts to earth; normally carries no current but carries fault current when necessary |

Operating current normally flows through live and neutral. The earth conductor provides protection during a fault.

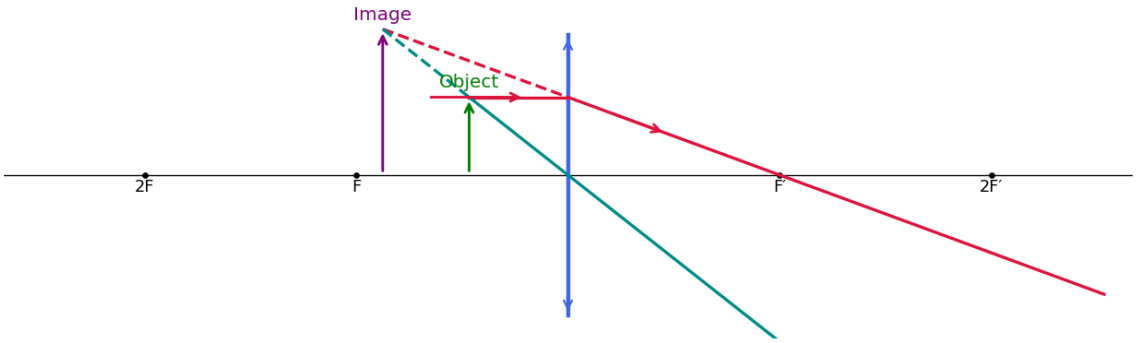
# Appendix A. Lens and eye ray diagrams

## A1-A2. Real and virtual images - P-10-D-47 and P-10-D-50

A1. Real image: object beyond 2F




A2. Virtual image: object inside F

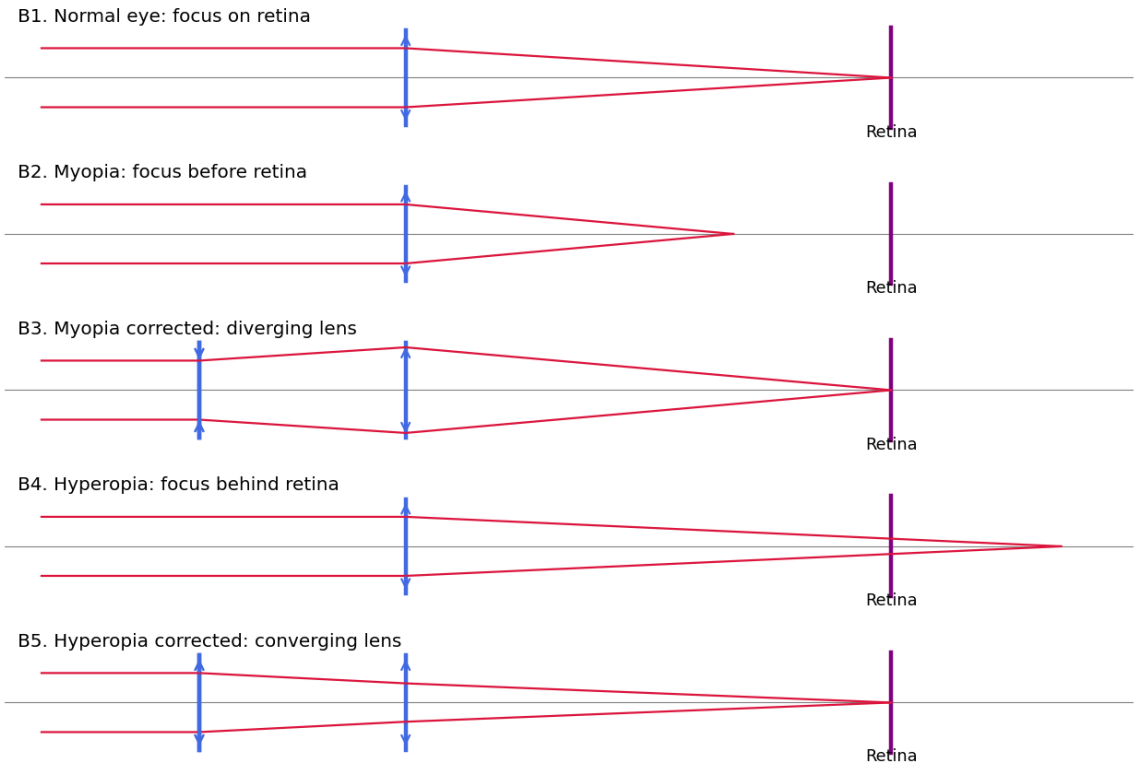



### Construction

Draw a ray parallel to the principal axis and refract it through the far focus. Draw a second ray through the optical centre. For a real image, the refracted rays meet. For a virtual image, extend them backwards using dashed lines.

Label the object, image, optical centre, principal axis, F and 2F. Add ray-direction arrows. The virtual image is upright and lies on the object side.

# A3. Normal and corrected eyes




## P-10-D-54 and P-10-D-55

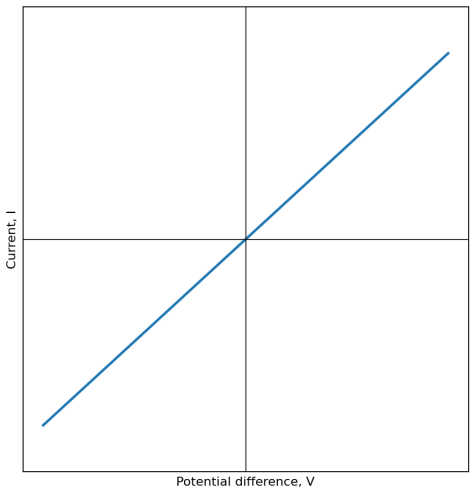
Normal: focus on the retina. Myopia: distant light focuses before the retina; a diverging lens corrects it. Hyperopia: insufficient convergence would place the focus behind the retina; a converging lens adds optical power.

These are schematic convergence diagrams. For near objects, show diverging incident rays; an uncorrected hyperopic eye may be unable to accommodate sufficiently to focus them on the retina.

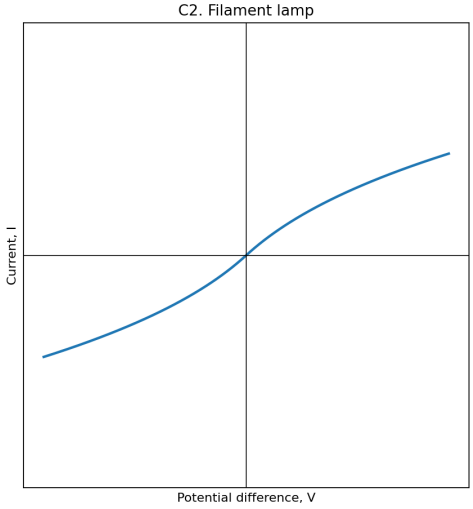
# Appendix B. Current-voltage graphs

P-10-E-36 | Horizontal axis: voltage. Vertical axis: current.

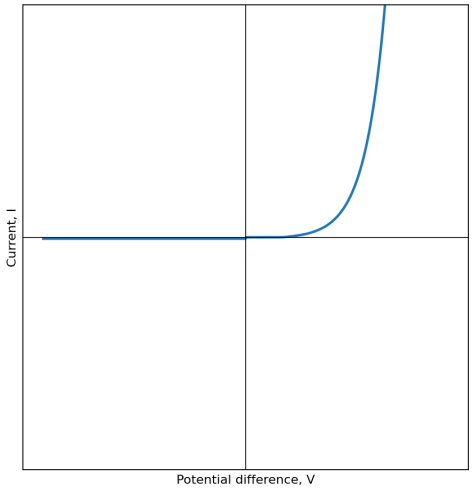
C1. Ohmic resistor




C2. Filament lamp



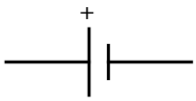

C3. Silicon diode



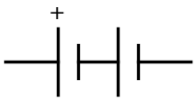
# Appendix C. Circuit symbols

P-10-E-37 | Standard school-circuit symbols and connections.

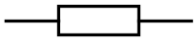
Cell



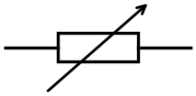

Battery



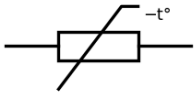

Fixed resistor



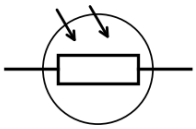
Variable resistor



NTC thermistor



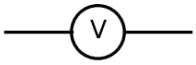

LDR



Ammeter




Voltmeter



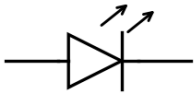

Open switch



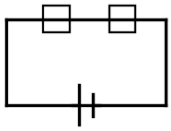
Closed switch



LED

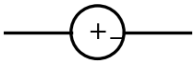


Series circuit



# Circuit symbols - continued

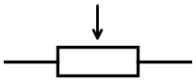
DC power supply




AC generator




Potential divider



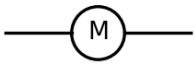
Heater



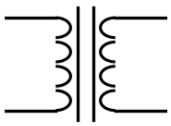
Lamp



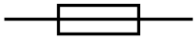
Motor



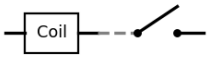

Transformer



Fuse



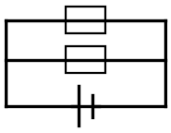
Relay



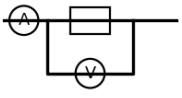

Diode



Parallel circuit



Meter connections




Ammeter: series connection. Voltmeter: parallel connection. LED arrows point outwards; LDR arrows point inwards.